

Dalhousie University
CSCI 3110 — Design and Analysis of Algorithms I
Fall 2024
Assignment 6

Distributed Tuesday, October 29 2024.

Due 11:59PM, Tuesday, November 5 2024.

Instructions:

1. Before starting to work on the assignment, make sure that you have read and understood policies regarding the assignments of this course, especially the policy regarding collaboration. <https://dal.brightspace.com/d2l/le/content/338900/Home?itemIdentifier=D2L.LE.Content.ContentObject.ModuleC0-4457275>
2. To submit via Crowdmark, upload a **separate** image/PDF file (multiple pages are allowed) to each question. You may resubmit as often as necessary until the due date. A good strategy is to create an initial submission days in advance after you solve some problems, and make resubmissions later. More detailed submission information for Crowdmark can be found at:
<https://crowdmark.com/help/completing-and-submitting-an-assignment/>
3. If you submit a joint assignment, use Crowdmark to create a group and add group members before making submissions. The “What Will Students See” section of <https://crowdmark.com/help/creating-a-group-assignment> shows the steps and screenshots of creating a group and adding members. Note that, “to avoid overwriting submissions”, Crowdmark recommends “only one group member submits all the work”.
4. We encourage you to typeset your solutions using LaTeX. However, you are free to use other software or submit scanned handwritten assignments as long as **they are legible**. We have the right to take marks off for illegible answers.

Questions:

1. [10 marks] Suppose you are the captain of a large, motor-propelled yacht sailing from Halifax, Canada to Buenos Aires, Argentina. Along the way, you plan to stop at n sea ports for business or pleasure, and port i is x_i nautical miles of sailing distance away from Halifax (port 0 is Halifax, while port n is Buenos Aires). You start your voyage at Halifax with a full fuel tank.

Let m be the number of nautical miles the yacht can sail with a full fuel tank. Your yacht is large, but it is not a superyacht, so m is far less than the total sailing distance.

This means you need to refuel at some of these sea ports. On the other hand, $m \geq x_i - x_{i-1}$ for any $i \in [1, n]$, and each of these n ports has a fuel terminal, so it is possible to arrive at the destination.

Each time you refuel, you need to stop by the fuel terminal and wait for your turn. Unfortunately, many of these sea ports are busy, and the time it takes for you to fuel your yacht can sometimes be long. Therefore, for each port i , there is an estimated fueling time t_i (mostly depending on how busy it is) given in the input, should you choose to refuel there.

Now, you would like to choose a subset of ports where you should refuel your yacht to minimize the total estimated fueling time, while still completing the voyage. The fueling time at the start (Halifax) or the destination (Buenos Aires) is not counted. You can assume that, each time you stop at one of these terminals, you always fill up your fuel tank.

It is easy to think of the following greedy strategy: each time you have a full fuel tank at a port, the next sea port where you refuel is the one that is the farthest away from your current port among all the subsequent ports that are within m miles. That is, if you fully refuel at port i (including the case where $i = 0$), then the next port to refuel is port j where j is the largest integer in $(i, n]$ with $x_j - x_i \leq m$.

Your first task is to give a counterexample to show that this greedy strategy will not always give us an optimal solution. Argue why the counterexample you construct is indeed a counterexample.

2. [10 marks] Now, use dynamic programming to design an efficient algorithm to solve the problem in question 1 optimally. In this question, you are only required to report the minimum total estimated fueling time.

The running time should be expressed as the big-Oh of a function of n , and m cannot be a variable of the function.

3. [10 marks] Further design an efficient algorithm to report the list of ports where you refuel in an optimal solution.